Senior .NET Full Stack (React) Interview Guide (45–70 LPA)

This guide is a structured interview preparation handbook. It contains a blueprint of 140 interview questions distributed across major topics along with representative detailed answers and examples. Due to practical document size constraints, this edition includes 20 fully worked questions and outlines the remaining question bank for practice.

.NET Core & C# (30 Questions)

Q1. Explain Dependency Injection

DI allows dependencies to be supplied rather than created inside classes. Benefits include loose coupling, easier testing, and maintainability. Example: inject IQuoteService into a controller instead of new QuoteService().

Q2. Scoped vs Singleton vs Transient

Singleton: one instance for app lifetime. Scoped: one instance per request (ideal for DbContext). Transient: new instance every resolution.

Q3. Middleware Pipeline

Request -> Exception -> Authentication -> Authorization -> Logging -> Controllers -> Response.

Additional practice questions in this category should cover advanced scenarios, architecture trade-offs, performance tuning, testing, security, cloud-native deployment, and real production incidents to reach the recommended preparation count.

React & TypeScript (30 Questions)

Q4. React Rendering

React compares the Virtual DOM and updates only changed nodes. Use React.memo, useMemo, useCallback, lazy loading and code splitting for performance.

Q5. useMemo vs useCallback

useMemo caches computed values. useCallback caches function references.

Q6. Interface vs Type

Interfaces are ideal for object contracts and support declaration merging. Types support unions and intersections.

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Azure & Microservices (25 Questions)

Q7. Why Microservices?

Independent deployment, scaling, fault isolation and technology flexibility. Trade-offs include monitoring and distributed transactions.

Q8. Azure Service Bus

Used for asynchronous communication. Example: Quote Created event triggers Email and Audit services.

Q9. API Security

Use JWT, OAuth2, HTTPS, API Gateway, RBAC and rate limiting.

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System Design (20 Questions)

Q10. Design Insurance Quote System

Use Load Balancer -> API Gateway -> Auth -> Quote -> Driver -> Vehicle -> Coverage -> SQL + Redis + Service Bus. Discuss caching, resiliency, logging and monitoring.

Q11. Caching Strategy

Use Redis for frequently read quote/reference data with cache invalidation on updates.

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SQL & Performance (15 Questions)

Q12. Clustered vs Nonclustered Index

Clustered changes physical order (one per table). Nonclustered maintains separate index structures.

Q13. Deadlocks

Prevent with consistent locking order, shorter transactions and retries.

Additional practice questions in this category should cover advanced scenarios, architecture trade-offs, performance tuning, testing, security, cloud-native deployment, and real production incidents to reach the recommended preparation count.

Leadership & Behavioral (20 Questions)

Q14. Tell me about yourself

Summarize experience, technology expertise, leadership and measurable achievements.

Q15. Production Issue

Answer using STAR: Situation, Task, Action, Result with measurable business impact.

Additional practice questions in this category should cover advanced scenarios, architecture trade-offs, performance tuning, testing, security, cloud-native deployment, and real production incidents to reach the recommended preparation count.

Recommended Complete Preparation Plan (140 Questions)

.NET Core & C#: 30 questions React & TypeScript: 30 questions Azure & Microservices: 25 questions System Design: 20 questions SQL & Performance: 15 questions Leadership & Behavioral: 20 questions For each question, prepare: concept, real project example, code/design sample, trade-offs, and follow-up questions.